

1. Core Concepts & Modelling

ITI state-space:

ẋ = Ax + Bu + D, y(t) = Cx

A=dynamics (main), B=input, C=output, D=disturbance. A ∈ ℝ<sup>n×n</sup>, B ∈ ℝ<sup>n×1</sup>, C ∈ ℝ<sup>1×n</sup>. Feedthrough form: y=Cx+Du.

ODE → SS: for ẍ + aẋ + bx = u, x = (x, ẋ)<sup>T</sup>:

A = (0 1; -b -a), B = (0; 1), C = (1 0)

3rd order ẍ + aẍ + bẋ + cx = u, x = (x, ẋ, ẍ)<sup>T</sup>:

A = (0 1 0; 0 0 1; -c -b -a), B = (0; 0; 1), C = (1 0 0)

TF → SS: for G(s) = b0 / (s^2 + a1s + a0):

A = (0 1; -a0 -a1), B = (0; 1), C = (b0 0)

3rd order G(s) = b0 / (s^3 + a2s^2 + a1s + a0):

A = (0 1 0; 0 0 1; -a0 -a1 -a2), B = (0; 0; 1), C = (b0 0 0)

Mass-spring-damper: mÿ + cẏ + ky = u, G(s) = 1 / (ms^2 + cs + k).

2. Linearisation (Lec 2)

For ẋ = f(x, u):
(1) Equilibrium f(xe, ue) = 0.
(2) Taylor (1st order):

ẋ ≈ f(xe, ue) + ∂f/∂x|e (x - xe) + ∂f/∂u|e (u - ue)

General (scalar) form: F(v0+δv) ≈ F(v0) + ∂F/∂v|v0 (v - v0).

(3) Deviation vars δx = x - xe, δu = u - ue:

δẋ ≈ ∂f/∂x|e δx + ∂f/∂u|e δu

Multivariable: A = ∂f/∂x|e (Jacobian), B = ∂f/∂u|e.

3. First-Order Systems (Lec 3)

ẋ + ax = u, free resp.: x(t) = e<sup>-at</sup>x(0).
Stability: pole at s = -a. a>0 ⇒ decays, stable; a<0 ⇒ grows, unstable.
Step (u=const), x(0)=x0: (don't forget step u · 1(t))

x(t) = (x0 - u/a) e<sup>-at</sup> + u/a

Steady state: x(∞) = u/a. Time constant τ = 1/a.
Time specs: rise tr ≈ 2.2/a = 2.2τ (10-90%); settle ts ≈ 3.91/a ≈ 4/a = 4τ (2%).

Transfer function: G(s) = output/input = Y(s)/U(s), y = G(s)u, u = Ue<sup>st</sup>.

G(s) = X(s)/U(s). For ẍ + 3ẋ + 2x = u: G = 1 / (s^2 + 3s + 2).

DC gain → final value: for any stable G, step of size U:
x(∞) = G(0) · U (set all derivs = 0).

G(0) sign = direction (G(0)<0 ⇒ output dips below 0); magnitude = final value. Generalises x∞=u/a.

4. Proportional / Steady-State Error

Plant ẋ + ax = u, u = KP(r - x) (state-feedback controller):

ẋ + (a + KP)x = KP r

x(∞) = KP r / (KP + a), error e(∞) = r · a / (KP + a).

With disturbance d, noise n:

xss = (KP(r - n) + d) / (a + KP)

Large KP ⇒ small error (feedback stabilisation: pick KP > -a).
Integral action ⇒ zero steady-state error.

SS error (unity fb, unit step): e(∞) = 1 / (1 + G(0)) (G(0)=DC loop gain; position const KP=G(0)).

System type: # poles of L=CG at s=0: 0, 1, 2 ⇒ Type 0, 1, 2.

|        | Type 0       | Type 1 | Type 2 |
|--------|--------------|--------|--------|
| Step   | 1 / (1 + KP) | 0      | 0      |
| Ramp   | ∞            | 1 / Kv | 0      |
| Parab. | ∞            | ∞      | 1 / Ka |

5. Homogeneous Solns & Poles (Lec 4)

Char. eq. of ẍ + aẋ + bx = 0: λ^2 + aλ + b = 0.
Real distinct: (a^2 - 4b > 0): xh = C1e<sup>λ1t</sup> + C2e<sup>λ2t</sup>.
Real repeated: (a^2 - 4b = 0): xh = (C1 + C2t)e<sup>λ\*t</sup>.
Complex: λ = -σ ± jωd:

xh = e<sup>-σt</sup> (C3 cos ωdt + C4 sin ωdt)

Stability: all poles (roots of denom.) have negative real part ⇔ x(t) → 0 for any x(0), zero input.
Necessary cond.: char. poly has all coefficients present & positive (same sign) — any missing/sign change ⇒ unstable. Real, distinct, negative poles ⇒ stable & non-oscillatory.

G(s) = (b1s + b0) / (s^2 + a1s + a0); poles=denom roots, zeros=numer roots.

6. Second-Order Systems (Lec 5)

Standard: G(s) = ωn^2 / (s^2 + 2ζωns + ωn^2).

Poles s = -ζωn ± jωn√(1 - ζ^2) = -σ ± jωd.

σ = ζωn, ωd = ωn√(1 - ζ^2)
ωn = √(σ^2 + ωd^2), ζ = σ/ωn

ζ = sine of angle from imag. axis. Underdamped: 0 < ζ < 1.

Envelope: x(t) = e<sup>-σt</sup> sin(ωdt).

Response specs: Mp = e<sup>-πζ/√(1-ζ^2)</sup> (overshoot), tp = π/ωd (peak time),

ts ≈ 4/ζωn (2% settle), tr ≈ 1.8/ωn (rise 10-90%).

7. PID Control (Lec 5)

C(s) = Kp + Ki/s + Kds = (Kds^2 + Kps + Ki)/s

- P: fast, may leave SS error / oscillate.
- PI: removes SS error, slower transient.

- PD: adds damping, faster, noise-sensitive, adds a zero.
  - PID: removes SS error + improves transient.
- Kp ↑ ⇒ ωn ↑ (faster), ζ ↓ (more overshoot).
Rate feedback: u = KP(r - x) - KDẋ (avoids derivative kick):

G(s) = (KP + KDs) / (s^2 + (1 + KD)s + KP)

zero at s = -KP/KD.
PD design ex. ẍ + 3ẋ + 2x = u, u = KP(r - x) - Kdẋ:

G = Kp / (s^2 + (3 + Kd)s + (Kp + 2))

match to s^2 + 2ζωns + ωn^2.

9. State-Space Response (Lec 6)

Matrix exponential: e<sup>At</sup> = ∑ (At)^k / k! = I + At + (At)^2 / 2! + ...

Free resp.: x(t) = e<sup>At</sup>x(0).

Forced: x(t) = e<sup>At</sup>x0 + ∫0^t e<sup>A(t-τ)</sup>Bu dτ.

TF from SS: G(s) = C(sI - A)<sup>-1</sup>B + D.
Output to u = Ue<sup>st</sup>: y = Ce<sup>At</sup>x0 + G(s)Ue<sup>st</sup>.
Stable ⇔ all eigenvalues of A have Re < 0.
e<sup>At</sup> = Te<sup>Jt</sup>T<sup>-1</sup> (Jordan); repeated eig give t<sup>k</sup> terms.

10. Reachability / Control (Lec 7)

Reachability matrix: (3rd order: 3 × 3)
W = [B AB A^2B]

Reachable ⇔ det W ≠ 0.
Equivalent: can place any closed-loop poles via K, for K = (k1 k2).
State feedback: u = -Kx, closed loop ẋ = (A - BK)x.

Design: det(sI - (A - BK)) <sup>1</sup> = desired char. poly, match coefficients.
Transformation: T = WW<sup>-1</sup> (W from reachable canonical form), K = K̃T.

Reachable canonical form: for ÿ + a1ẏ + a2y = u (top-row companion):
Ã = (-a1 1; -a2 0), B̃ = (1 0)

char. poly s^2 + a1s + a2 read off top row.
Dominant poles: choose 2nd-order pair for transient; place rest 5-10× further left, low overshoot.

11. Observability / Observers (Lec 8)

Observability matrix: (3rd order: 3 × 3)

Wo = [C; CA; CA^2]

Observable ⇔ det Wo ≠ 0: recover x(0) from u, y on [0, T].
Luenberger observer:

ẋ̂ = Aẋ̂ + Bu + L(y - Cẋ̂)

Error x̃ = x - ẋ̂: ẋ̃ = (A - LC)x̃.
Condition: A - LC stable (⇒ x̃ → 0). Roots (eig) of A - LC: negative ⇒ error decays, x̃ → x (convergent); positive ⇒ unstable, error grows, never converges.

Design: det(sI - (A - LC)) <sup>1</sup> = desired poly, match coeffs.
Output feedback: (separation): observer + u = -Kẋ̂ + Nr. CL char poly: det(sI - (A - BK)) det(sI - (A - LC)).
Separation principle: systems both reachable & observable: design cont K and obsv L independently; the CL poles are just the union of

the controller poles (eig  $A-BK$ ) and observer poles (eig  $A-LC$ ).

**Feedforward:** (zero SS err):  $N = -(C(A-BK)^{-1}B)^{-1}$ .

**Integral action:** augment state  $\dot{q} = Cx - \bar{y}$  to kill SS error without exact model.

12. Frequency Response (Lec 9)

Input  $A \sin \omega t \Rightarrow y_{ss} = A|G(j\omega)| \sin(\omega t + \phi)$

$G(j\omega) = |G(j\omega)|e^{j\phi}$ ,  $\phi = \angle G = \arctan \frac{\text{Im}}{\text{Re}}$ .

$\angle G = \sum \angle(\text{num}) - \sum \angle(\text{den})$ : **numerator** (zeros) add +ve phase, **denominator** (poles) add -ve phase.

**Complex algebra:**  $z_1 z_2$ : multiply mags, add phases;  $z_1/z_2$ : divide mags, subtract phases.

Ex.  $G = \frac{1}{j\omega(j\omega+1)(j\omega+3)}$ :

$$|G| = \frac{1}{\sqrt{\omega^2(\omega^2+1)(\omega^2+9)}}$$

$$\angle G = -(\frac{\pi}{2} + \arctan \omega + \arctan \frac{\omega}{3})$$

**Forced sinusoid (undetermined coeffs):** for  $\ddot{x} + \omega_n^2 x = F \sin \omega t$ , try  $x = X \sin \omega t$ , match:

$$X = \frac{F}{\omega_n^2 - \omega^2}$$

real  $\Rightarrow$  phase  $0^\circ/180^\circ$  (no lag);  $X \rightarrow \infty$  as  $\omega \rightarrow \omega_n$  (**resonance**, undamped).

13. Bode Plots (Lec 9–10)

Mag in dB:  $20 \log_{10} |G(j\omega)|$ .  $K = 10^{dB/20}$ ,  $dB = 20 \log_{10} K$ .  $\times 10 \Rightarrow 20\text{dB}$ ;  $1 \rightarrow 0\text{dB}$ . **Standard form:** factor so each bracket starts with 1; constant out front = DC gain  $K$ :

$$G(s) = \frac{K(\frac{s}{a}+1)}{s^n(\frac{s}{b}+1)}$$

**Start height:**  $H(j\omega)_{\text{dB}} = 20 \log_{10}(K\omega) = 20 \log_{10}(K) + 20 \log_{10}(\omega)$

**Mag slopes:** (added at break freq  $\omega=|p|, |z|$ ):

|                             |                                                                           |
|-----------------------------|---------------------------------------------------------------------------|
| DC gain $K$                 | flat at $20 \log K$                                                       |
| integrator $1/s^n$          | $-20n$ dB/dec from $\omega \rightarrow 0$                                 |
| diff. $s$                   | $+20$ dB/dec                                                              |
| pole $(\frac{s}{b}+1)^{-1}$ | $-20$ dB/dec after $\omega=b$                                             |
| zero $(\frac{s}{a}+1)$      | $+20$ dB/dec after $\omega=a$                                             |
| cplx poles @ $\omega_n$     | $-40$ dB/dec; peak $M_z = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$ at $\omega_n$ |
| cplx zeros @ $\omega_n$     | $+40$ dB/dec                                                              |

**Phase:**  $\angle G = \sum \angle(\text{num}) - \sum \angle(\text{den})$ ; each  $(j\omega+a)$ :  $\arctan \frac{\omega}{a}$ ; transi-

|                         |                                |
|-------------------------|--------------------------------|
| DC gain                 | $0^\circ$                      |
| integrator $1/s$        | $-90^\circ$ (all $\omega$ )    |
| diff. $s$               | $+90^\circ$ (all $\omega$ )    |
| pole @ $b$              | $-90^\circ$ total, centred $b$ |
| zero @ $a$              | $+90^\circ$ total, centred $a$ |
| cplx poles @ $\omega_n$ | $-180^\circ$ total             |

**Bode  $\rightarrow$  TF:** **(1)** DC gain  $K=10^{X\text{dB}/20}$  from flat low- $\omega$  ( $\omega = 1$  if slanted start). **(2)** initial slope: flat=no int.,  $-20/\text{dec}$ =one  $1/s$ ,  $-40=1/s^2$ ,  $+20=s$ . **(3)** corners where slope *changes*: extra  $-20 \Rightarrow$  pole,  $+20 \Rightarrow$  zero,  $\mp 40 \Rightarrow$  cplx pair @  $\omega_n$ . **(4)** phase confirms: start  $0/-90/-180^\circ=0/1/2$  integrators; drop @ corner=pole, rise=zero. **(5)** assemble:

$$G(s) = \frac{K s^{\text{pos}}(s+z_1)(s+z_2)\cdots}{s^{\text{int}}(s+p_1)(s+p_2)\cdots}$$

**Sanity:** final slope =  $-20(\#p-\#z)$  dB/dec; final phase =  $-90^\circ(\#p-\#z)$ ; DC check: sub  $s=0$ .

**GM & PM on Bode:** GM = magnitude (dB) at phase =  $-180^\circ$ ; PM = phase (deg) at magnitude 0 dB.

14. Nyquist Criterion (Lec 10,12)

Plot  $L(j\omega)$ ,  $\omega: -\infty \rightarrow \infty$ . Closed loop  $G_{yr} = \frac{L}{1+L}$ .

**Theorem:**  $P$ =open-loop RHP poles of  $L$ ;  $N$ =net *clockwise* encirclements of  $-1$ . Then CL has  $Z = N + P$  RHP poles.

**Stable**  $\Leftrightarrow Z = 0 \Leftrightarrow N = -P$  ( $P$  anticlockwise encirclements).

Stable- $L$  case: stable iff no encirclements of  $-1$ .

CL poles satisfy  $L(s) = -1$ .

**Real-axis crossing:** Nyquist evaluates  $G$  along the imag. axis, so to find where it crosses  $(\alpha, 0)$ : set  $G(s)=\alpha$  and solve — the root is **purely imaginary**  $s = \pm j\omega$ .

15. Stability Margins (Lec 10,11)

Crossover  $\omega_{gc}$ :  $|L(j\omega_{gc})| = 1$  (0 dB).

**Gain margin:**  $g_m = 1/|L(j\omega_{pc})|$  at phase =  $-180^\circ$ . Find where Nyquist crosses -ve real axis; GM = inverse of that magnitude. (+ve).

**Phase margin:**  $\varphi_m = 180^\circ + \angle L(j\omega_{gc})$ . Find where Nyquist hits unit circle; PM = angle from that point to -ve  $x$ -axis.

**Stability margin:**  $s_m$ =closest dist of Nyquist to  $-1$ ;  $|S(j\omega)| = 1/\text{dist to } (-1)$ . Shortest distance from Nyquist curve to  $(-1, 0)$ .  $|S(j\omega)|$  = sensitivity.

$$PM = 2 \sin^{-1} \left( \frac{1}{2|S(j\omega_c)|} \right).$$

**Time delay:** Max delay  $\tau_{max} = \varphi_m/\omega_{gc}$ .

16. Sensitivity / Gang of Four (Lec 11)

$$S = \frac{1}{1+PC}, \quad T = \frac{PC}{1+PC}, \quad S+T=1$$

$S+T=1$ : a seesaw — low  $S$  (tracking/dist. rejection) vs. low  $T$  (robust to noise/model uncertainty); shape each small in different bands.

**3 TFs:**  $d, r, n.$   $\rightarrow$  Set other 2 to 0 and solve.

Disturbance:  $\eta = PSd$ .  $|S|<1$  attenuates;  $|S|=0$  eliminates;  $|S|>1$  worsens.

$S$  measures sensitivity of CL to plant uncertainty.

**Loop shaping — why:** shape  $L=PC$  to balance performance vs stability.

**how:** (i) high gain / integrator at low  $\omega \rightarrow$  tracking + reject  $d$  ( $S \approx 0$ ); (ii) low gain + steep roll-off at high  $\omega \rightarrow$  reject noise  $n$  ( $T \approx 0$ ); (iii) gentle slope ( $\approx -20\text{dB/dec}$ ) near  $\omega_{gc} \rightarrow$  keep phase margin.

Or pick desired  $L$  (e.g.  $\frac{\alpha}{s}$ ,  $\alpha$  sets  $\omega_{gc}$ ) then  $C=L/P$  — avoid cancelling unstable poles/zeros.

**Robustness:** relative model error  $\Delta m(s) = \frac{\Delta G(s)}{G(s)}$ ; robust stability

$$\Leftarrow |T(j\omega)| < \frac{1}{|\Delta m(j\omega)|} \text{ — keep } T \text{ small where } \Delta m \text{ large (high } \omega).$$

17. Fundamental Limitations (Lec 11,12)

**(1)**  $S+T=1$  — seesaw trade-off.

**(2)** Gain-phase relation — steep roll-off  $\Rightarrow$  phase lag, low PM.

**(3)** Interpolation constraints — RHP poles/zeros can't be hidden.

**(4)** Bode sensitivity integral (waterbed):

$$\int_0^\infty \log |S(j\omega)| d\omega = \pi \sum_k p_k$$

$p_k$ =unstable open-loop poles; push  $|S|$  down somewhere  $\Rightarrow$  up elsewhere.

19. Block Diagram Algebra

**Series:**  $G_1 G_2$ . **Parallel:**  $G_1 \pm G_2$ .

**Neg. feedback:**  $\frac{G}{1+GH}$ ; unity:  $\frac{G}{1+G}$ .

**Pos. feedback:**  $\frac{G}{1-GH}$ .

$L(s) = C(s)P(s)$  (loop TF),  $T(s) = \frac{L(s)}{1+L(s)}$ ,  $S(s) = \frac{1}{1+L(s)}$  (sensitivity function measures rel change of CL TF wrt to small change of the plant).

**Cascaded SS — recipe:** EOM of each subsystem:

$$\dot{x}_1 = A_1 x_1 + B_1 u_1, \quad y_1 = C_1 x_1; \quad \dot{x}_2 = A_2 x_2 + B_2 u_2, \quad y_2 = C_2 x_2$$

Sub interconnection  $u_2 = y_1 = C_1 x_1$  into  $\dot{x}_2$ ; stack  $x = [x_1; x_2]$ :

$$\dot{x} = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix} x + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u_1, \quad y_2 = [0 \quad C_2] x$$

**TF:** get  $G_i = C_i(sI - A_i)^{-1} B_i + D_i$  for each subsystem, then  $G_{tot} = G_2 G_1$  (multiply) — far easier than inverting the big  $(sI - A)$ .

20. Quick Reference / Exam Tips

- Stable  $\Leftrightarrow$  all poles/eigenvalues in open LHP.
- Reachable:  $\det[B \ AB] \neq 0$ ; Observable:  $\det[C \ CA] \neq 0$  (2nd order).
- PD design: pick  $\zeta$  from  $M_p$ ,  $\omega_n$  from  $t_s$ , match denom.
- Nyquist through  $(\alpha, 0)$ : set  $G(s) = \alpha$ ; root is purely imaginary  $s = \pm j\omega$ .
- Zero SS error tracking: use **integral** action (robust) or feedforward  $N$  (needs exact model).
- Real-world uncertainty: parameter errors, disturbances, sensor noise, noise affecting dynamics, unmodelled nonlinearity.

**PID vs state-feedback:** **PID:** model-free, mainly SISO, heuristic/robust, simple to tune, limited pole control. **State-FB:** needs accurate model + full state (or observer), handles MIMO, gives exact pole placement.

**Useful:**  $\zeta=0.5 \Rightarrow M_p \approx 16\%$ ;  $\zeta=0.707 \Rightarrow M_p \approx 4.3\%$ ;  $\zeta=0.6 \Rightarrow M_p \approx 9.5\%$ .

$\sqrt{1-\zeta^2}$  at  $\zeta=0.707$  is 0.707.

21. Written Question Framework

**Observability:** Can uniquely recover initial value  $x(0)$  from measured  $u, y$  over  $[0, T]$ . Find unmeasurable from what you can measure.

**Observer purpose:** estimate unmeasured state  $x$  from available  $u, y \rightarrow$  enables state-feedback when full state unavailable. Asymptotically tracks state of the plant.

**Sensitivity function  $S$ :** measures system sensitivity to plant uncertainty;  $|S| \ll 1$  means disturbances attenuated (good tracking, disturbance rejection).  $S+T=1$  (seesaw): cannot make both small.

**$S$  &  $T$  trade-off:**  $S$  low at low  $\omega$  (track, reject slow disturbances);  $T$  low at high  $\omega$  (reject fast noise). Near crossover  $\omega_{gc}$ , phase margin enforces balance.

**Gain margin:** the factor by which the loop gain can be increased before closed-loop instability occurs. Larger GM  $\Rightarrow$  more robust to gain errors/uncertainty.

**Real-world control design:** (1) model system (ODE/SS from physics); (2) ID parameters (experiments, data); (3) check reachable/observable; (4) design  $K$  for transient specs ( $\zeta, \omega_n$ ); (5) design  $L$  for measurement noise tolerance; (6) check closed-loop stability (Nyquist/poles); (7) simulate with real noise/uncertainty.

**Step response features from TF:**  $G(s)$  poles  $\Rightarrow$  poles in TF  $\Rightarrow$  envelope decay  $e^{-\sigma t}$  and oscillation frequency  $\omega_d$ . Final value  $x(\infty) = G(0) \cdot U$ . DC gain sign  $\Rightarrow$  direction (dip/rise).

22. Math Toolbox

$$2 \times 2 \text{ inverse: } \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

$$3 \times 3 \text{ det: } \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei-fh) - b(di-fg) + c(dh-eg).$$

$$2 \times 2 \text{ eigenvalues: } \det(\lambda I - A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$$

$$\text{Common integrals: } \int e^{-at} dt = -\frac{1}{a} e^{-at}; \int_0^\infty e^{-at} dt = \frac{1}{a} \quad (a>0).$$

$$\text{Sin/cos deriv: } \frac{d}{dt} \sin \omega t = \omega \cos \omega t, \quad \frac{d}{dt} \cos \omega t = -\omega \sin \omega t.$$

$$\text{Sin/cos integ: } \int \sin \omega t dt = -\frac{1}{\omega} \cos \omega t, \quad \int \cos \omega t dt = \frac{1}{\omega} \sin \omega t.$$